Part I: Analogy

Instruction: Answer questions (1 - 12) based on the pair that best expresses relationship similar to that of the original or the capitalized pair. Then, click the button of your choice.

1. THIGH : LEG

(A) palm : hand

(B) thumb : finger

(C) tie : neck

(D) arm : shoulder

2. DILIGENCE : SUCCESS

(A) excitement : stimulation

(B) victory : ecstasy

(C) vivacity : animation

(D) triumph : achievement

3. FISH : SARDINE

(A) herd : cattle

(B) camel : hump

(C) cow : calf

(D) fowl : chicken

4. PARENT : CHILD

(A) couple : offspring

(B) father : mother

(C) brother : sister

(D) mother : cousin

5. DERMATOLOGY : SKIN

(A) logic : reasoning

(B) histology : tissue

(C) chronology : crime

(D) etymology : society

6. PERCEPTABLE

(A) noticeable

(B) trivial

(C) undetectable

(D) visible

7. OFFENCE

(A) congruence

(B) virtue

(C) immorality

(D) cruelty

8. SERENE

(A) bright

(B) noisy

(C) sedentary

(D) quiet

9. PROHIBIT

(A) encourage

(B) deter

(C) connect

(D) block

Part III: Sentence Completion

Instruction: Each of the following sentences (23 - 32) consists of one or two blank spaces indicating that one or two words have been omitted. Choose from the alternatives given the word or words which best complete each sentence.

10. There are several reasons why school uniforms must be _____ for all students.

(A) worn

(B) sought

(C) mandatory

(D) found

11. Computer _____ can be defined as the excessive or compulsive use of the computer which persists despite serious negative _____.

(A) browsing ... functions

(B) addiction ... consequences

(C) processing ... applications

(D) programming ... outlooks

12. Almost every aspect of modern life has been _____ through convenience provided by technology.

(A) affirmed

(B) improved

(C) disjointed

(D) connected

13. Unlike human beings, the computer _____ repetitive calculations without getting tired or _____.

(A) indicates ... observed

(B) classifies ... noticed

(C) demonstrates ... stuck

(D) performs ... bored

14. In order to succeed in class, students need to _____ the resources available to them throughout their studies.

(A) distinguish

(B) clarify

(C) utilize

(D) produce

15. It is certain that advertising is an everyday _____ of our lives.

(A) feature

(B) demand

(C) forecast

(D) discussion

16. In the strictest sense, the _____ of a material would produce a _____ supply of the original.

(A) discarding ... fewer

(B) creation ... larger

(C) extracting ... limitless

(D) recycling ... fresh

17. Obesity is a medical condition in which excess body fat is _____ to the extent that it may have an adverse effect on health.

(A) developed

(B) accumulated

(C) hardened

(D) risen

18. She always gets up early _____ be late for work.

(A) in order not to

(B) in order to

(C) in order

(D) in order that she

19. She asked me if I _____ the film before.

(A) had seen

(B) see

(C) have seen

(D) have had

20. The committee chose to postpone the meeting rather than _____ it that day.

(A) has

(B) having

(C) to have

(D) not having

21. We all left the place _____ as soon as we heard the blast.

(A) with hurry

(B) hurried

(C) hurriedly

(D) be in a hurry

22. She didn't want to stop and talk to me _____

(A) as she was very busy

(B) in spite of her busy schedule

(C) for I asked her

(D) irrespective of being busy

23. The title Grandmaster is awarded to chess players by the world chess organization FIDE. Apart from World Champion, Grandmaster is the highest title a chess player can attain. Once achieved, the title is held for life. According to this text, which of the following is true?

(A) World Champion is a title that is held for life.

(B) Grandmaster is a higher title than World Champion.

(C) World Champion is a higher title than Grandmaster.

(D) The title World Champion is awarded by the world chess organization FIDE.

24. The Mayan languages form a language family spoken in Meso-America and Northern Central America. Mayan languages are spoken by at least 6 million indigenous Maya, primarily in Guatemala, Mexico, Belize and Honduras. In 1996, Guatemala formally recognized 21 Mayan languages by name, and Mexico recognizes eight more. The Mayan language family is one of the best documented and most studied in the Americas. Modern Mayan languages descend from Proto-Mayan, a language thought to have been spoken at least 5,000 years ago. Based on the above text, which of the following statements is true?

(A) Proto-Mayan is a modern variation of Mayan.

(B) Not much is known about the Mayan language family.

(C) The Mayan language is the least studied language family.

(D) Mayan languages are mainly spoken in South America.

25. Six years ago, lawyer-banker-scholar Charles Morris wrote a prophetic book – 'Two-Trillion-Dollar Meltdown: Easy Money, High-Rollers and the Great Credit Crunch' – that foresaw the 2008 Great Recession before it clobbered America and the world. Now Morris has reversed course and sees good times ahead. His forthcoming book, 'Comeback,' predicts that surging U.S. energy independence will bring a buoyant rise in American manufacturing and jobs. This passage implies that you should pay attention to Charles Morris because:

(A) His predictions have come true once already.

(B) He's a published author.

(C) The U.S. is becoming more energy independent.

(D) His first book was a best seller.

26. Text I: Climate change - can we stop it?
It is implied in the second paragraph of the text that:

(A) climate change is caused by human actions.

(B) global warming is the result of climate change.

(C) people are doing their level best to combat global warming.

(D) human beings are not responsible for climate change.

27. The writer of the text believes that:

(A) all the energy that humans use will certainly come from renewable sources before the year 2100.

(B) it won't be long before people completely get rid of carbon as a source of energy.

(C) there will be more than enough energy for everyone from renewable sources before

2100.

(D) humans have a long way to go before they realize the benefits of renewable energy sources.

28. According to the writer of the text,

(A) it is pointless to try to stop climate change.

(B) there is debate about whether climate change is real or not.

(C) there is general consensus that climate change is real.

(D) climate change is real and there is nothing that can be done about it.

29. Which one of the following is true according to the text?

(A) The generation of heat and energy leads to global warming.

(B) It is not possible to become "carbon neutral" before 2050.

(C) The use of coal and oil results in the emission of carbon.

(D) It was decided at the Glasgow Summit to stop using coal and oil as of November 2021.

30. The phrase "other people" in the last paragraph refers to people who:

(A) make promises but do not keep them.

(B) are not aware of climate change.

(C) do not care about the world around them.

(D) are concerned about our planet.

31. Based on the information in the text, it can be said that:

(A) it is possible to combat global warming on condition that humans act in time.

(B) the green energy revolution is too slow to fight against climate catastrophe.

(C) it has become too late now to combat global warming.

(D) not many people feel the need to do something about global warming.

32. The writer of the text:

(A) believes that it is not possible to convince people that climate change is real.

(B) thinks that the green revolution is unlikely to solve the problems of climate change.

(C) disagrees that global warming is the fault of human beings.

(D) is critical of people who have priorities other than stopping climate catastrophe.

33. Which one of the following is NOT a renewable source of energy?

(A) electricity

(B) wind

(C) Sun

(D) sea

34. One disadvantage of renewable energy is that:

(A) a lot of people are reluctant to face the challenges of using it.

(B) some people believe that it is too costly to be used.

(C) some of the sources are unreliable.

(D) it requires the use of expensive technology.

35. The world is getting hotter because:

(A) of climate catastrophe.

(B) of climate change.

(C) people are not aware of it.

(D) there is only one Earth.

36. In the year 2000 E.C, the price of a jacket was Birr p. In each next year, the price increases by 20% of the price of the previous year, while in 2003, the price decreases by 10%. What was the total percentage increase in price of the jacket from year 2000 to 2003?

(A) 34%

(B) 30%

(C) 30.6%

(D) 29.6%

37. There are 29 students in the classroom, among which 15 are girls. The average weight of girls is 42 kg, and the average weight of boys is 45 kg. If a new student weighing 54 kg joins the classroom, what will be the average weight of all the students in the classroom in kilogram?

(A) 43.8

(B) 43.5

(C) 44

(D) 47

38. Which one of the following is true about the two quantities?

Quantity M	**Quantity N**
$(526)^2 - (525)^2$	210

(A) Quantity M is less than quantity N.

(B) Quantity M is greater than quantity N.

(C) Quantity M is equal to quantity N.

(D) The quantities cannot be determined.

39. The following graph shows the amounts of export and import (in million US$) of certain company per year from 2016 to 2020 E.C.
Which one of the following is true about from the graph the export and import of the company?

(A) The ratio of the amount of export to the amount of import in 2016 is 3:2.

(B) The ratio of the total export to the total import over the given period is 8:9.

(C) The maximum difference between export and import is $10 million each year.

(D) The export is increased by 90% from 2016 to 2020.

40. In the figure below, BE is a diameter of the circle, ABCD is a rectangle with area 72m^2, $CD = 8\text{m}$ and $DE = 3\text{m}$. How long is the radius of the circle in meter? (Figure not reproduced, but options given.)

(A) 5

(B) 4.5

(C) 4

(D) 5.5

41. The following bar graph represents the number of HIV positive people among five social groups, coded as SG1, SG2, SG3, SG4 and SG5, in certain town. Moreover, the adjacent table shows the total number of people in each of these social groups.
Which one of the following is true about the given data?

(A) 3% of those in SG4 are HIV positive.

(B) SG3 consists of the highest percentage of HIV positive people.

(C) SG1 consists of the highest percentage of HIV positive people.

(D) 5.8% of the entire population of the five social groups are HIV positive.

42. What is the least value of number x for which $(2x + 1)(x - 2) = 0$?

(A) 2

(B) $\frac{-1}{2}$

(C) -1

(D) -2

43. The shaded region R on the following coordinate plane is the intersection of $y \leq x + 2$ and $y \geq x^2$. If x is the first coordinate of any of the points (x, y) in the region R and y is the second coordinate of any of the points (x, y) in the same region R, what is the maximum possible value $2y - x$?

(A) 6

(B) 8

(C) 9

(D) 10

44. The following table presents a simple frequency distribution of scores of a variable.

Score	Frequency
2	**1**
5	**4**
6	**2**
8	**2**
10	**1**

Which one of the following is true about the mean, mode and median of this distribution?

(A) The median is the average of the mode and the mean.

(B) The mode is greater than the mean and the median.

(C) The mean, mode and median are equal.

(D) The mean is the average of the mode and the median.

45. Two positive quantities x and y are related by the equation $y = \frac{k}{x}$, where k is a constant. What will happen to the value of y if the value of x will increase by 50%?

(A) decrease by 66.7%

(B) decrease by 33.3%

(C) decrease by 50%

(D) increase by 50%

46. If $x = 10^{0.3} + \sqrt{0.017}$, $10^{0.6} = 4$, $\sqrt{1.7} = 1.3$, and $y = \sqrt{x^2 + 2.7}$, then which one of the following is the best approximate of y^2?

(A) 6.5

(B) 7.4

(C) 6.7

(D) 7.2

47. Last year a man spent between one-fifth and one-fourth of his yearly net income on his bank loan payments. If the man spent birr 45,000 on his bank loan payments last year, which one of the following could have been his yearly net income (in birr) last year?

(A) 170,000

(B) 228,000

(C) 222,000

(D) 178,000

48. Which number is NOT in the sequence: 2, 6, 12, 20, 32, 42, 56, ...?

(A) 32

(B) 56

(C) 42

(D) 72

49. Which one of the following is simplified form of $\sqrt{\frac{539}{242}} \div \sqrt{\frac{189}{66}}$?

(A) $\frac{1}{7}\sqrt{6}$

(B) $\frac{1}{7}\sqrt{3}$

(C) $\frac{1}{3}\sqrt{14}$

(D) $\frac{1}{3}\sqrt{7}$

50. If $a - b = p$, then which one of the following is the value of $a^2 - 2ab + b^2$?

(A) $p^2 + ap - b^2$

(B) $2p$

(C) p^2

(D) $p(a - b)$

51. The circle below has center O and the perimeter of $\triangle AOB$ is 12 units. Which of the following is NOT true? (Figure implied but not reproduced.)

(A) Area of $\triangle AOB$ is $4\sqrt{3}$ sq. units.

(B) The area of the circle is 12π sq. units.

(C) AOB is isosceles.

(D) The perimeter of the circle is 8π units.

52. $4x - 2y = 8$, then what is the value of $2x - y$? (No options provided in text; likely a free-response but here we treat as implicit. However, from PDF no options given; we skip option list. But we preserve the question for completeness.)

[**Note: Question 76: $4x - 2y = 8$, then $2x - y =?$ — no answer choices visible**]

53. What is the value of m in the sequence $\frac{1}{2}, \frac{3}{5}, \frac{8}{11}, \frac{19}{23}, \frac{42}{47}, m, \ldots$?

(A) $\frac{89}{95}$

(B) $\frac{76}{91}$

(C) $\frac{85}{97}$

(D) $\frac{65}{87}$

54. If $n = 3$, then which one of the following the value of $2^{2n} + 1$?

(A) 65

(B) 64

(C) 33

(D) 129

55. The following table presents a simple frequency distribution of scores of a variable.

Score	Frequency
2	**1**
5	**4**
6	**2**
8	**2**
10	**1**

What is the standard deviation of the distribution?

(A) $\sqrt{4.4}$

(B) $\sqrt{5.4}$

(C) $\sqrt{6}$

(D) $\sqrt{5}$

56. Statement: More than 95% of the people working in Ethiotelecom know programming.

(A) More than 5% of the people working in Ethiotelecom don't know programming.

(B) Less than 5% of the people working in Ethiotelecom don't know programming.

(C) 5% of the people working in Ethiotelecom do not know anything other than programming.

(D) 5% of the people working in Ethiotelecom know something other than programming.

57. The government has allowed unsubsidized educational institutions to increase their fees.

(A) Subsidized educational institutions need to increase their fees.

(B) Unsubsidized educational institutions have less money to improve their infrastructure.

(C) Subsidized educational institutions do not need to increase their fees.

(D) Unsubsidized educational institutions are established by rich people.

58. It is vital that we increase public spending on healthcare to keep the population in good health.

(A) Public spending on healthcare is too low.

(B) Public spending on healthcare is too high.

(C) The health of the population is in decline.

(D) Public healthcare is the concern of this Government.

59. In Ethiopian context, the advisable age for a child to join a kindergarten is 6 years.

(A) At this age, the child is ready to learn new things.

(B) Kindergartens do not take admission of children who are more than 6 years old.

(C) After this age, kids do not like to go to kindergarten.

(D) The children not enrolled in Kindergarten do not perform well.